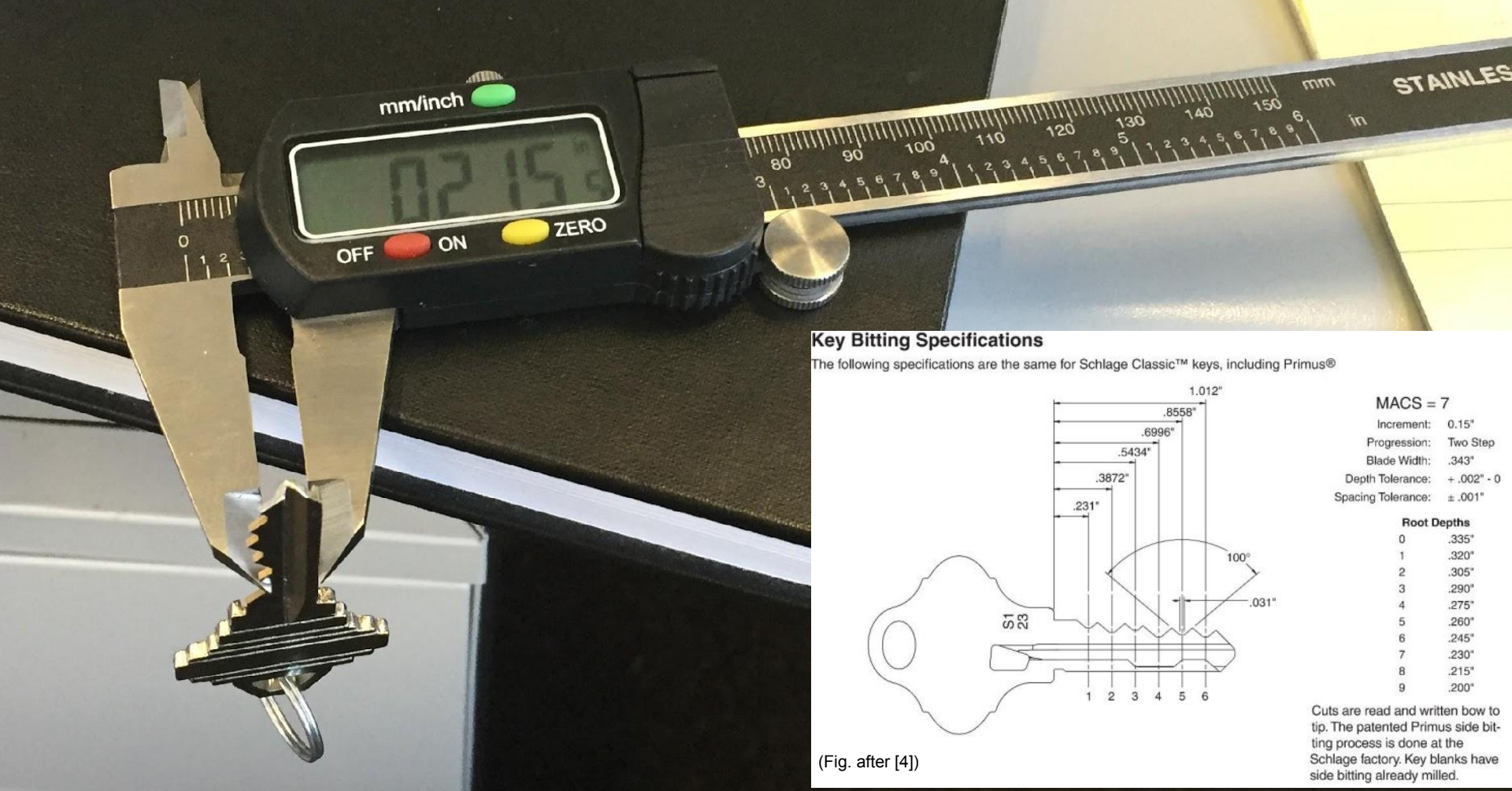


Key Bitting Specifications

The following specifications are the same for Schlage Classic™ keys, including Primus®

MACS = 7

Increment:	0.15"
Progression:	Two Step
Blade Width:	.343"
Depth Tolerance:	+ .002" - 0
Spacing Tolerance:	± .001"

Root Depths

0	.335"
1	.320"
2	.305"
3	.290"
4	.275"
5	.260"
6	.245"
7	.230"
8	.215"
9	.200"

Cuts are read and written bow to tip. The patented Primus side bitting process is done at the Schlage factory. Key blanks have side bitting already milled.

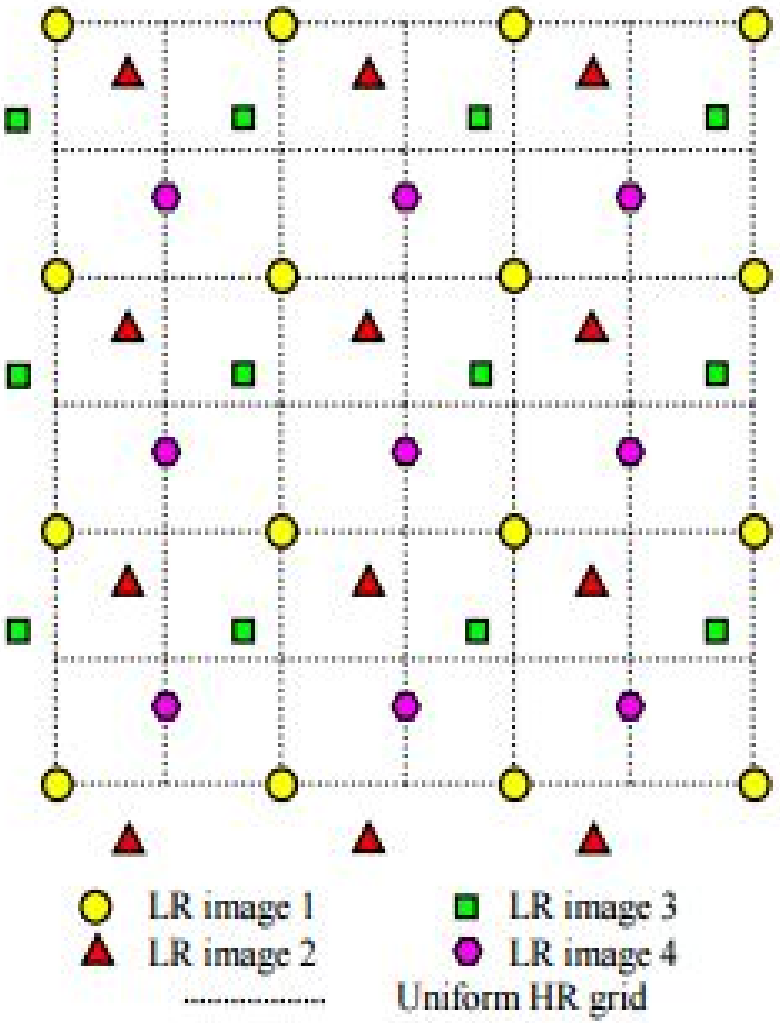
(Fig. after [4])

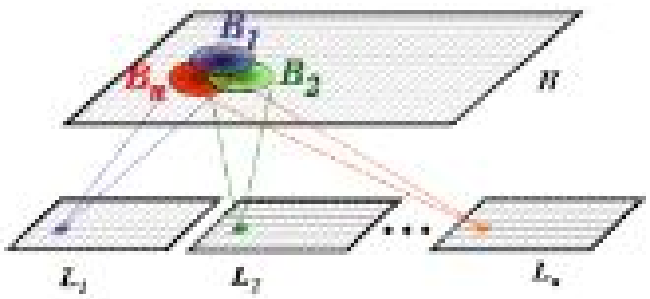
Z S H C
H S K R N
C H K R V D
H O N S D C V
O K H D N R C S
V H D N K U O S R C
B D C L K Z V H S R O A
H K O B C A N O M P V E S R

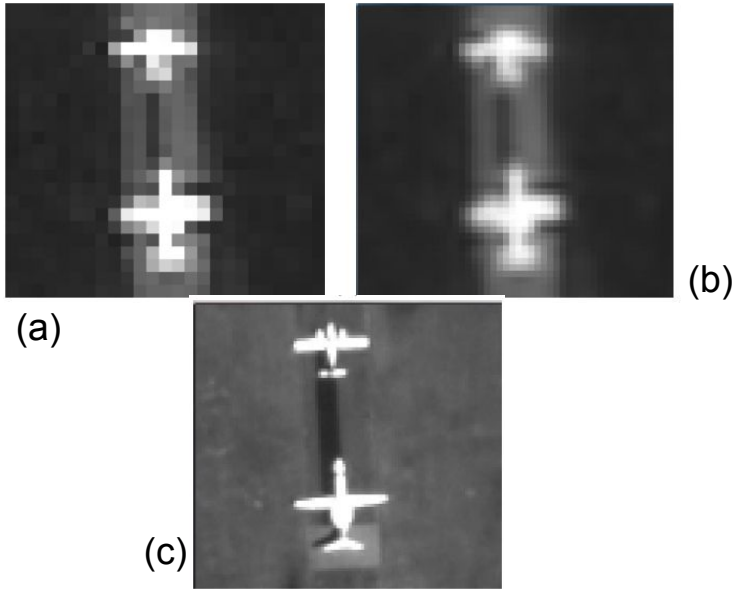
Z S H C
H S K R N
C H K R V D
H O N S D C V
O K H D N R C S
V H D N K U O S R C
B D C L K Z V H S R O A
H K O B C A N O M P V E S R

(Fig. after [7])

Left: [6], Upper right: [7], Bottom right: [9]

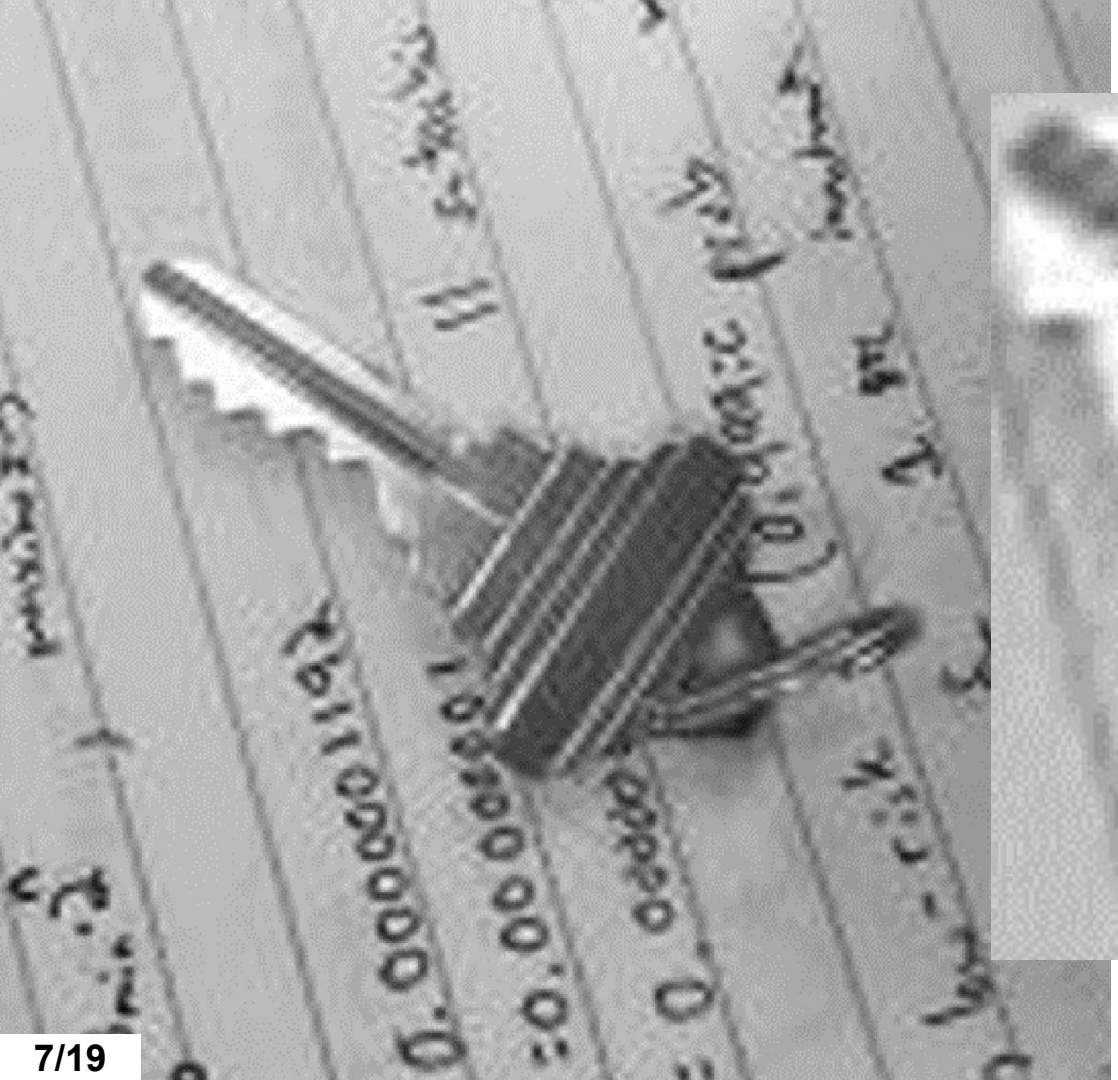


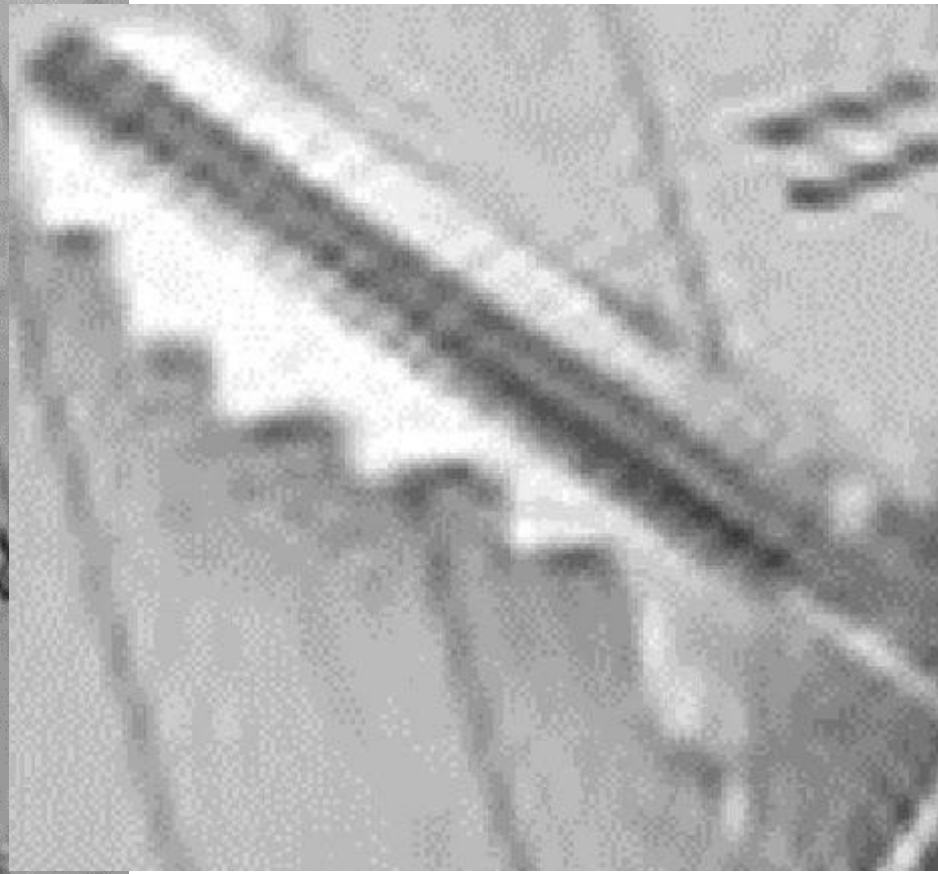


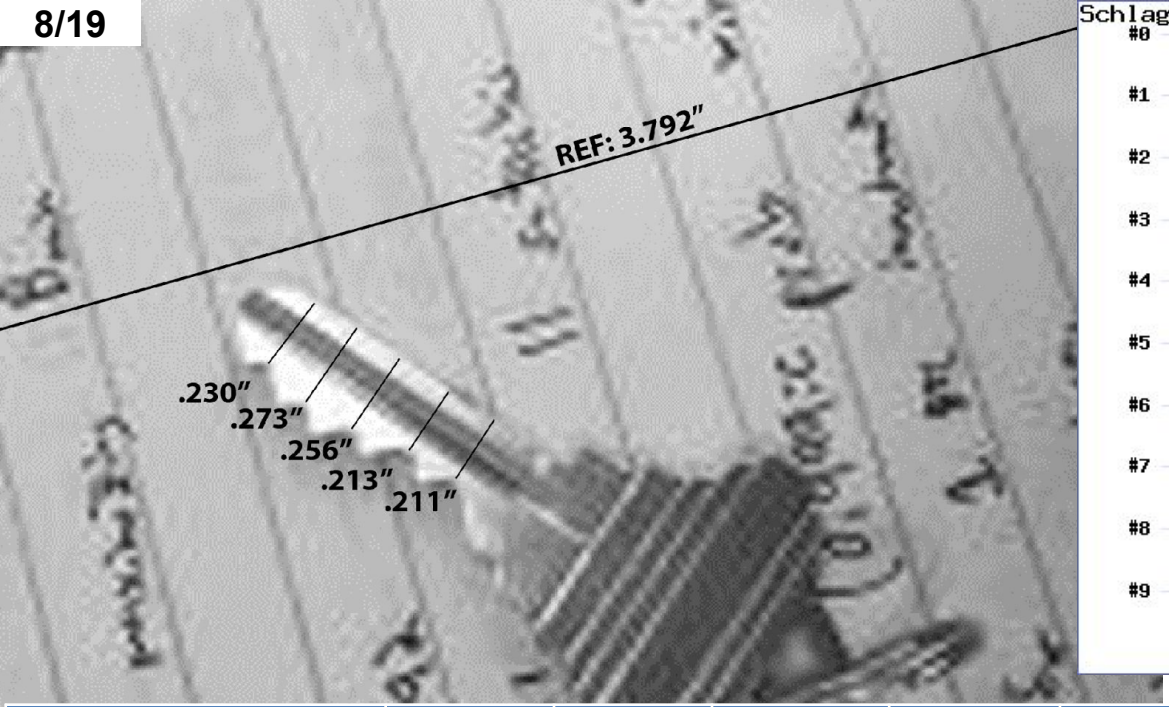


Cubic Spline Interpolation on best image (at 180x240)

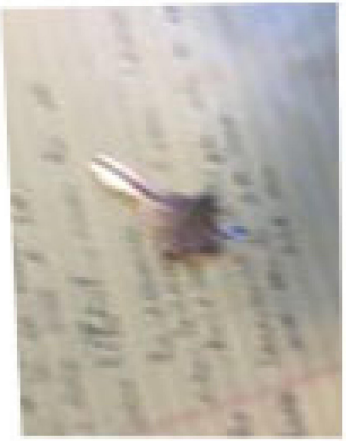
“Fast and Robust Multiframe Super Resolution” on all 112 images (at 90x120)

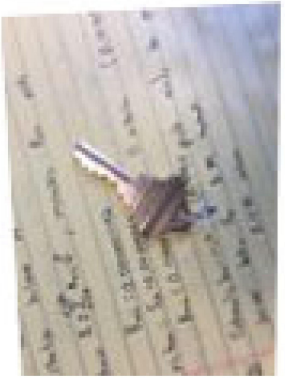


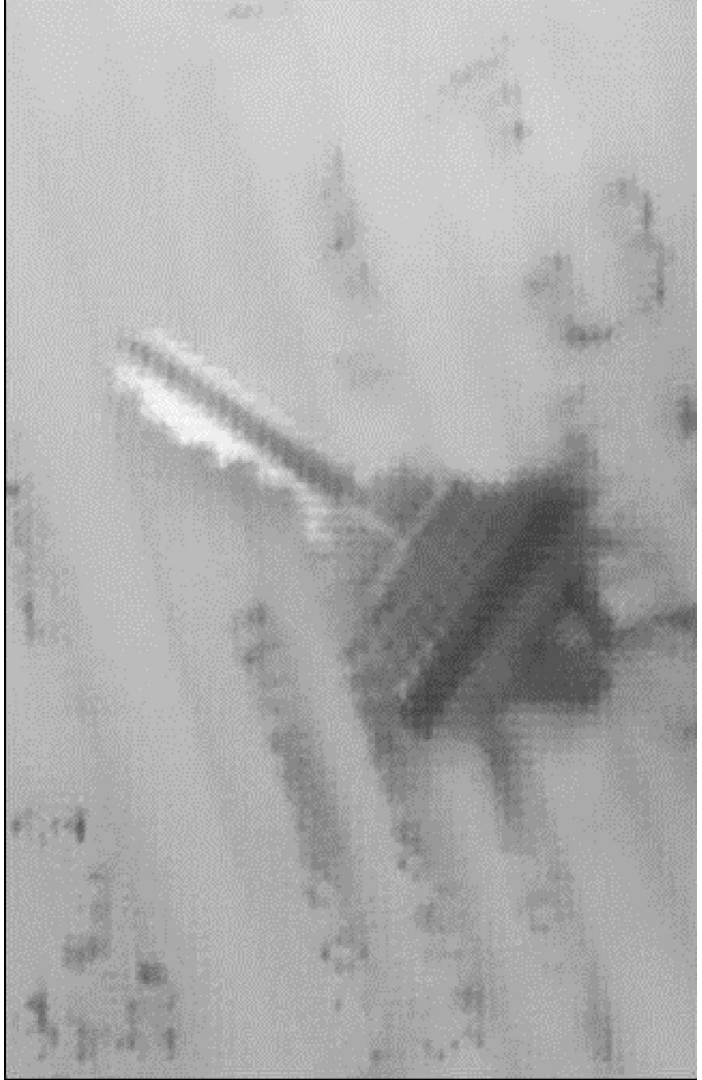


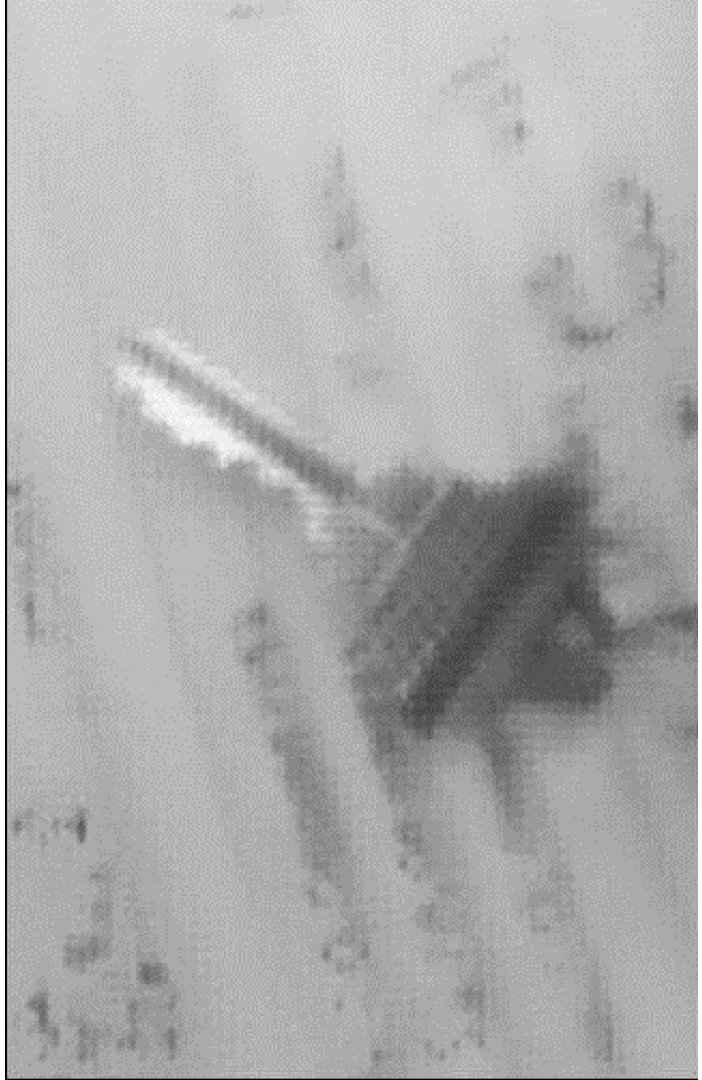


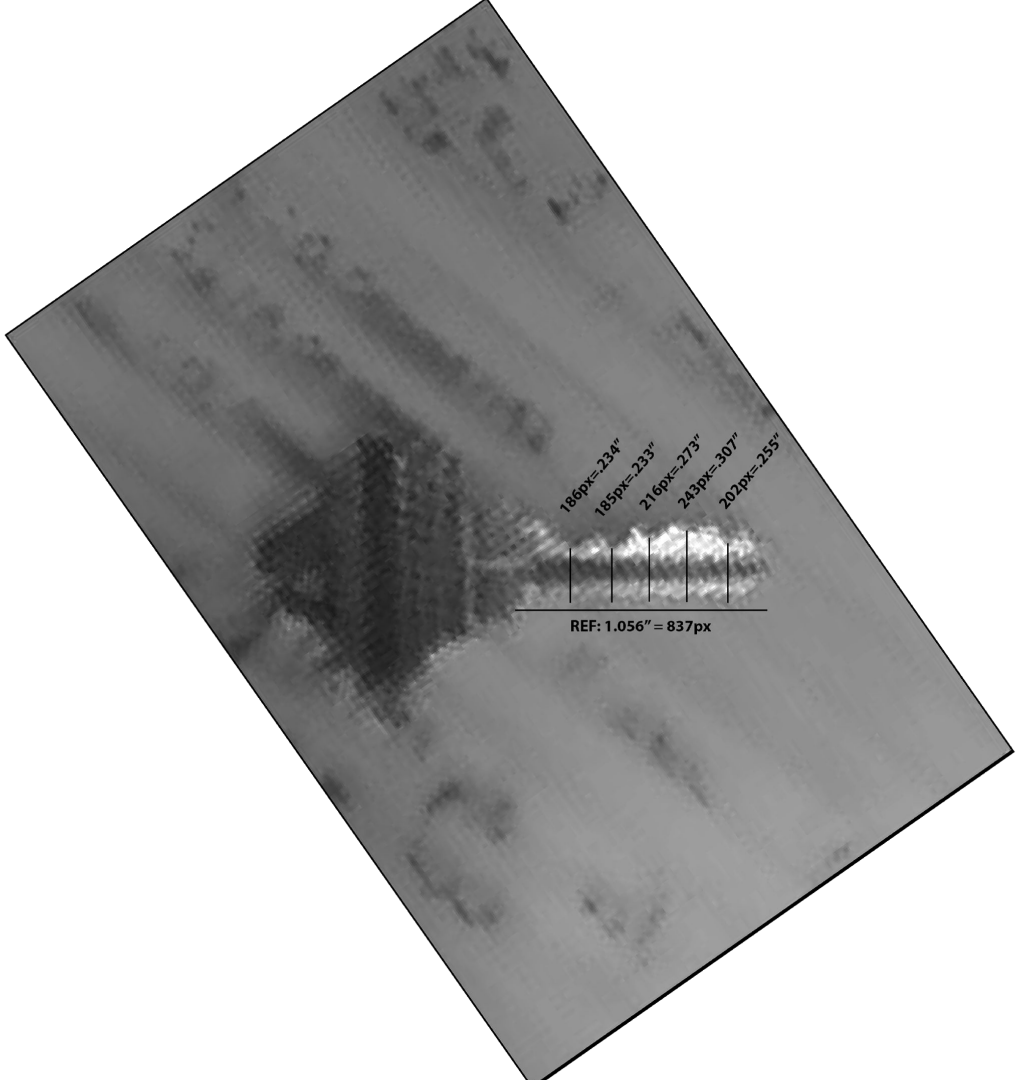
Pin	1	2	3	4	5
Measured width	0.211”	0.213”	0.256”	0.273”	0.230”
Actual width	0.215”	0.215”	0.250”	0.270”	0.232”
Measured code	8	8	6	4	7
Actual code	8	8	5	4	7

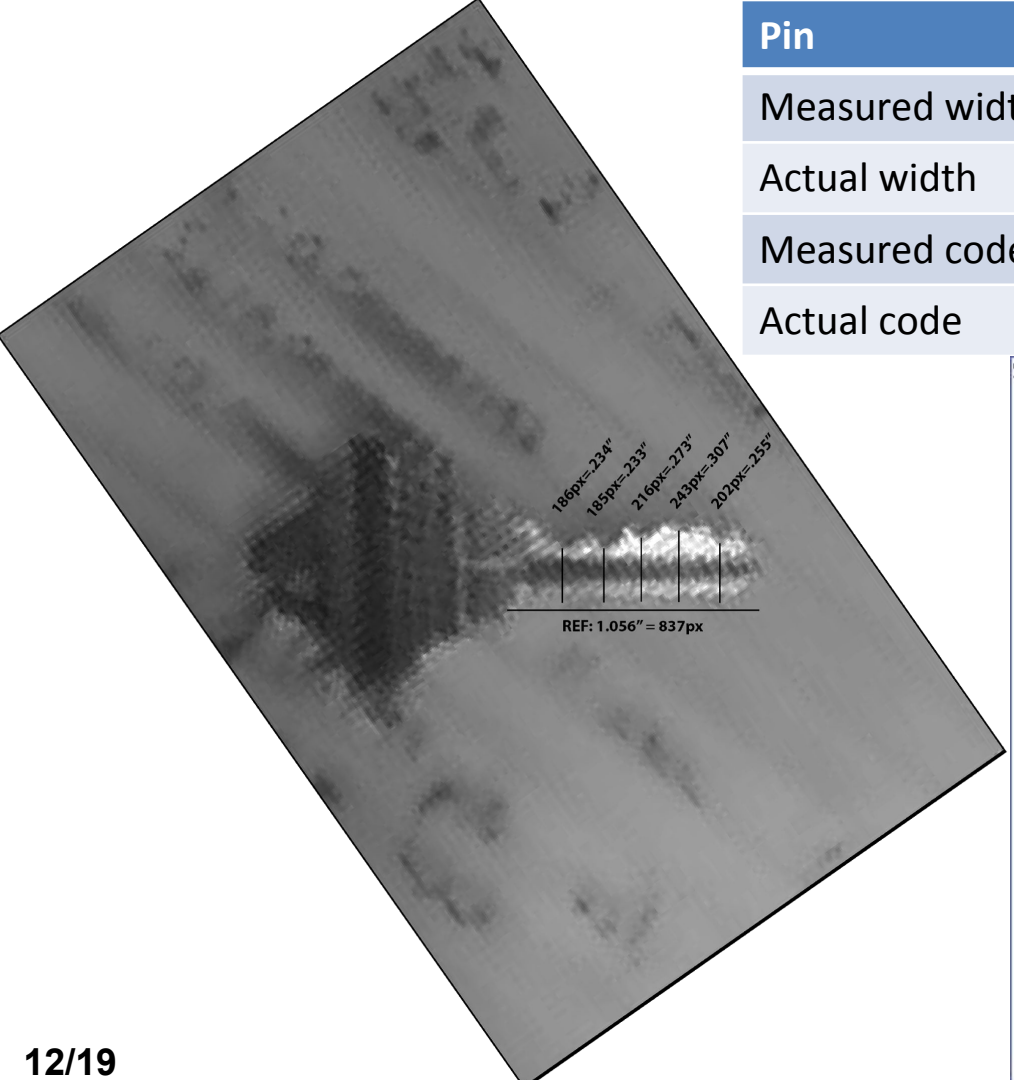









186px=.234"
185px=.233"
216px=.273"
243px=.307"
202px=.255"
REF: 1.056" = 837px



Pin	1	2	3	4	5
Measured width	0.234”	0.233”	0.273”	0.307”	0.255”
Actual width	0.215”	0.215”	0.250”	0.270”	0.232”
Measured code	**7**	**7**	**4**	**2**	**5**
Actual code	8	8	5	4	7

http://iank.org/312.pdf

[1] Farsiu, S., Robinson, M.D., Elad, M. and Milanfar, P., 2004. Fast and robust multiframe super resolution. *Image processing, IEEE Transactions on*,*13*(10), pp.1327-1344.
[2] Bouguet, J.Y., 2001. Pyramidal implementation of the affine lucas kanade feature tracker description of the algorithm. *Intel Corporation*, *5*(1-10), p.4.
[3] Hanson, O. Robust and fast super resolution. http://www1.idc.ac.il/toky/videoProc-07/projects/SuperRes/srproject.html. Access April 2016.
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[6] Gilman, A., Bailey, D.G. and Marsland, S.R., 2008, January. Interpolation models for image super-resolution. In *Electronic Design, Test and Applications, 2008. DELTA 2008. 4th IEEE International Symposium on* (pp. 55-60). IEEE.
[7] Glasner, Daniel, Shai Bagon, and Michal Irani. "Super-resolution from a single image." In Computer Vision, 2009 IEEE 12th International Conference on, pp. 349-356. IEEE, 2009.
[8] B. Laxton, K. Wang, and S. Savage. Reconsidering physical key secrecy: Teleduplication via optical decoding. In Proceedings of the 15th ACM conference on Computer and communications security, pages 469–478. ACM, 2008.
[9] Zhang, Hongyan, Zeyu Yang, Liangpei Zhang, and Huanfeng Shen. "Super-resolution reconstruction for multi-angle remote sensing images considering resolution differences." Remote Sensing 6, no. 1 (2014): 637-657.

- Structure from motion for high security keys
- Multi-sensor camera systems
- Motion registration w/ inertial data
- Probabilistic decoding
- Perspective correction
- Compromised smartphones / IP cams / etc
- app

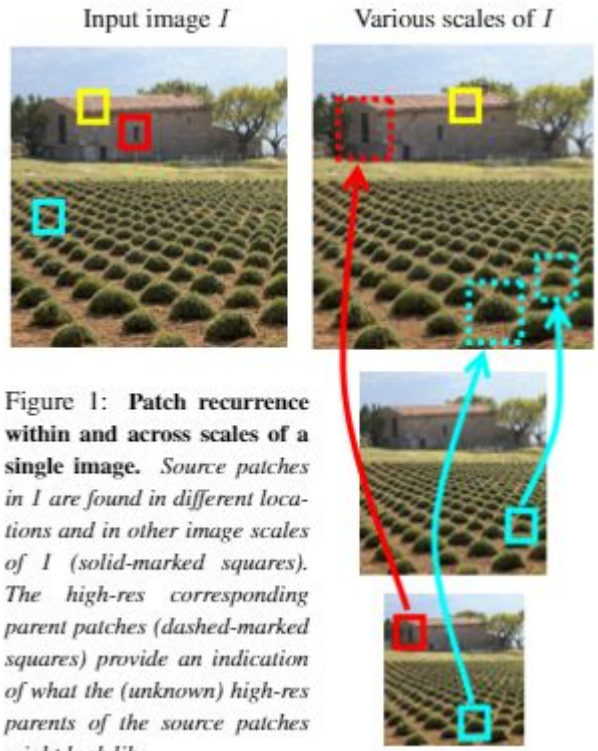
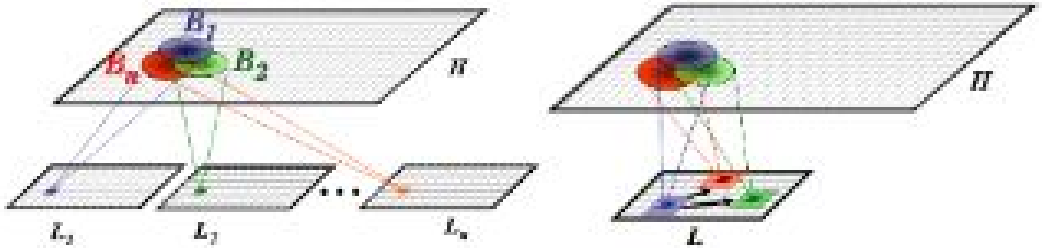
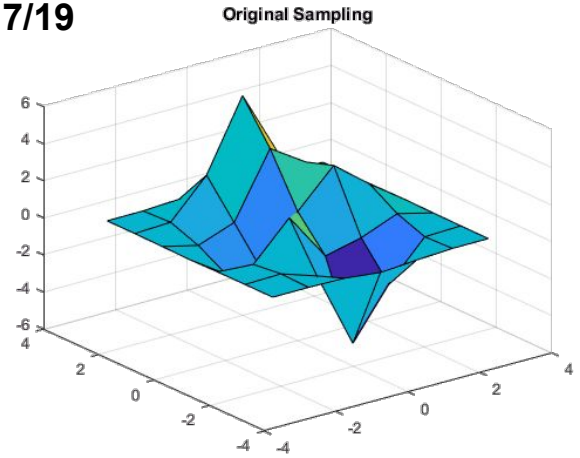


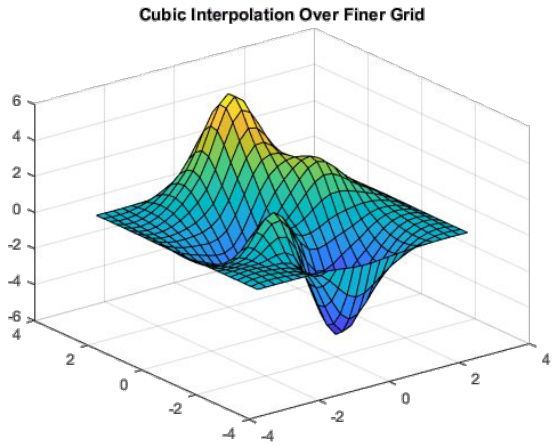
Figure 1: **Patch recurrence within and across scales of a single image.** *Source patches in I are found in different locations and in other image scales of I (solid-marked squares). The high-res corresponding parent patches (dashed-marked squares) provide an indication of what the (unknown) high-res parents of the source patches might look like.*

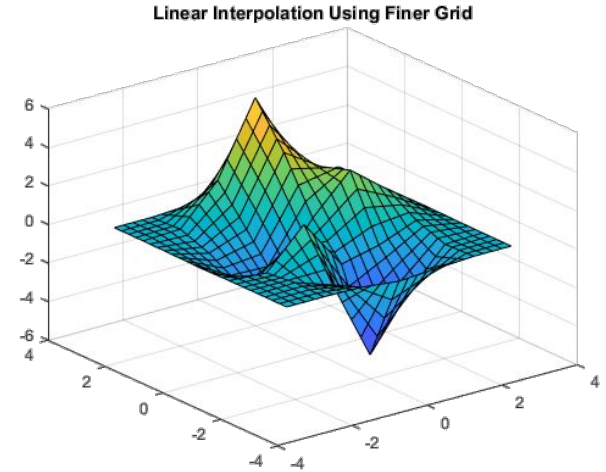


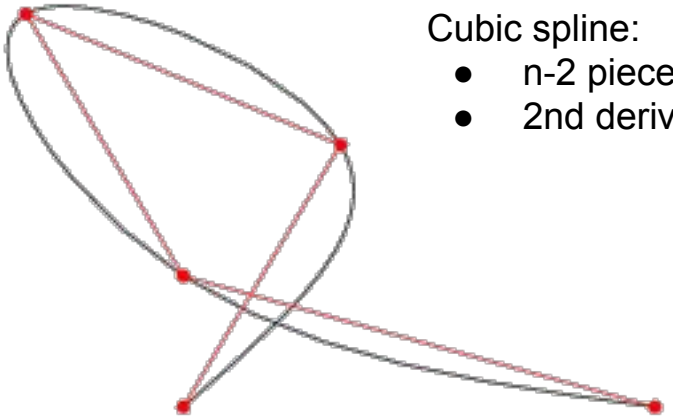
(a) Classical Multi-Image SR (b) Single-Image Multi-Patch SR

Glasner, Daniel, Shai Bagon, and Michal Irani. "Super-resolution from a single image." In Computer Vision, 2009 IEEE 12th International Conference on, pp. 349-356. IEEE, 2009.









Cubic spline:

- n-2 piecewise 3rd-order polynomials
- 2nd derivative at endpoints = 0

Fig 1,2,3 https://www.mathworks.com/help/matlab/ref/interp2.html.
Fig 4 http://mathworld.wolfram.com/CubicSpline.html

$$\Upsilon_{TV}(\underline{X}) = \|\nabla \underline{X}\|_1$$

$$\Upsilon_{BTV}(X) = \sum_{l=0}^{P} \sum_{m=0}^{P} \alpha^{m+l} \|\underline{X} - S_x^l S_y^m \underline{X}\|_1$$

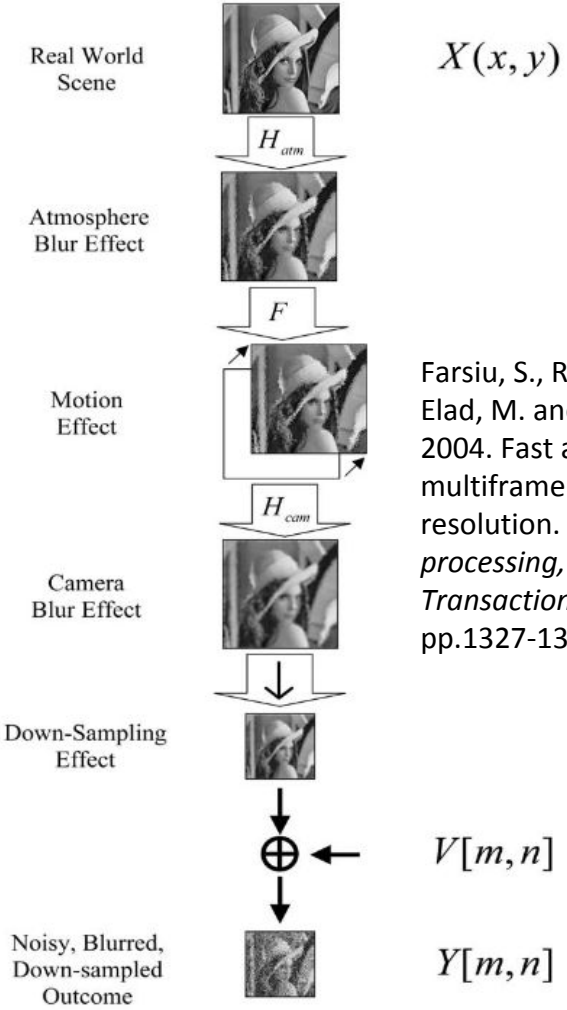
Farsiu, S., Robinson, M.D., Elad, M. and Milanfar, P., 2004. Fast and robust multiframe super resolution. *Image processing, IEEE Transactions on*,*13*(10), pp.1327-1344.

These functions penalize variation

Given corrupted image Y, reconstruct X by minimizing:

$$\underline{\widehat{X}} = \underset{\underline{X}}{\text{ArgMin}} \left[\|\underline{Y} - \underline{X}\|_2^2 + \lambda \Upsilon(\underline{X}) \right]$$

i.e., estimate X should be similar to Y, but with a reduction in variance controlled by parameter lambda



Farsiu, S., Robinson, M.D., Elad, M. and Milanfar, P., 2004. Fast and robust multiframe super resolution. *Image processing, IEEE Transactions on*,*13*(10), pp.1327-1344.

F, H, D, S: motion, blur, decimation, shift
2nd term: regularization (similarity cost)
Minimize errors in X while penalizing variation

$$\underline{\hat{X}} = \underset{\underline{X}}{\text{ArgMin}} \left[\sum_{k=1}^{N} \|D_k H_k F_k \underline{X} - \underline{Y}_k\|_1 + \lambda \sum_{l=0}^{P} \sum_{m=0}^{P} \alpha^{m+l} \|\underline{X} - S_x^l S_y^m \underline{X}\|_1 \right]$$

Solve using steepest descent:

$$\underline{\hat{X}}_{n+1} = \underline{\hat{X}}_n - \beta \left\{ \sum_{k=1}^{N} F_k^T H_k^T D_k^T sign(D_k H_k F_k \underline{\hat{X}}_n - \underline{Y}_k) + \lambda \sum_{l=0}^{P} \sum_{m=0}^{P} \alpha^{m+l} [I - S_y^{-m} S_x^{-l}] sign(\underline{\hat{X}}_n - S_x^l S_y^m \underline{\hat{X}}_n) \right\}$$